Mohith Kumar Atmakur

Data Engineer

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OBJECTIVE

Experienced Senior Data Engineer with 5+ years of delivering scalable, secure, and high-performance data platforms across **AWS, GCP**, and **Azure**. Led end-to-end cloud **migrations**, real-time **ingestion**, and predictive **ML** pipelines using **Kafka**, **PySpark**, **Snowflake**, **BigQuery**, and **Databricks** driving 40–70% improvements in data efficiency, latency, and insight delivery. Proven impact across **healthcare, retail, and fintech** through regulatory-compliant (**HIPAA, GDPR**) architectures, robust **CI/CD** automation, and multi-source integrations supporting data science, fraud detection, and business intelligence.

WORK EXPERIENCE

FirstEnergy, Akron, Ohio, USA

Role: Azure Data Engineer — **Aug 2024 – Present**

PROJECT: At FirstEnergy, I focused on migrating high-volume energy consumption, equipment monitoring, and grid performance data from Netezza to Azure Synapse, building scalable cloud pipelines to support operational analytics, real-time outage reporting, and regulatory compliance across utility systems.

Responsibilities:

- Migrated **5–6 TB of customer data** from Netezza to Azure Synapse, modernizing FirstEnergy's data warehouse with **Medallion Architecture** and boosting reporting speed and SLA compliance by **40%**.
- Modelled **Bronze-Silver-Gold layers** with **Delta Lake tables in Synapse and Databricks**, increasing data quality to **99.9%** through modular PySpark/T-SQL transformations and UTC standardization.
- **Managed and optimized relational and NoSQL databases**, including sources like **Snowflake**, SQL Server, PostgreSQL, MySQL, Cassandra, and MongoDB, to support both structured and semi-structured financial data.
- Streamlined schema design, indexing, and partitioning strategies, which improved transactional query performance and storage efficiency by **30%** across enterprise reporting systems.
- **Designed optimized SQL queries** and applied dimensional modeling (star/snowflake schemas) to enhance OLAP and OLTP performance across MS SQL Server environments, ensuring consistent analytics at scale.
- Implemented real-time data ingestion using Kafka, Flume, and Kinesis to support **ML models** in Azure ML Studio, accelerating fraud detection and forecasting by 40%.
- Improved processing efficiency by 35% with PySpark, Hive, MapReduce, and advanced analysis using **Pandas** and **NumPy**.
- **Automated CI/CD pipelines** with Git, Maven, Azure DevOps, and Apache Airflow, achieving **99.8% uptime** for ETL workflows and reducing deployment rollback frequency through robust QA integration.
- Designed and deployed Azure infrastructure components using **Terraform**, including Data Factory pipelines and Synapse workspaces for orchestrating enterprise data workflows.
- Collaborated with business teams to build modular and scalable ELT models using **DBT** for Snowflake transformations, enhancing maintainability and auditability.
- **Developed Power BI dashboards** with complex DAX metrics and PowerShell automation, improving executive visibility into store-level sales, costs, and margin KPIs in near real time.
- **Enhanced query performance** and storage efficiency on large-scale datasets using Impala, Trino, Sqoop, and **stored procedures**, lowering financial report generation latency by **45%**.
- **Built secure web applications** with Flask and Django for internal analytics teams, using Linux shell scripting on CentOS, and integrated automated testing and monitoring with Selenium, Cucumber, and JIRA, cutting support overhead by **35%**.
- Worked cross-functionally in **Agile sprints**, collaborating with analysts, ML engineers, and business stakeholders. Participated in sprint planning, peer code reviews, backlog grooming, and Jira-based reporting to ensure alignment and delivery.

ENVIRONMENT: Azure Synapse Analytics, Azure Data Lake Storage, Azure Data Factory, Python, PySpark, SQL, Netezza, Azure Monitor, Log Analytics, Shell Scripting, Git, Jenkins, Power BI.

Cleveland Clinic Health System, Cleveland, Ohio, USA

Role: AWS Data Engineer **Nov 2023 – Jun 2024**

PROJECT: At Cleveland Clinic Health System, I focused on designing scalable data architectures and cloud-based ETL workflows to support healthcare analytics and clinical operations. I ensured secure, compliant data pipelines (HIPAA) on AWS, collaborated with data scientists to enable research-driven insights, and maintained AWS infrastructure to support mission-critical applications, patient data flows.

Responsibilities:

- Designed and deployed scalable **ETL pipelines** using **AWS Glue**, **Lambda**, **Informatica**, and **Terraform** to process structured and semi-structured healthcare data across hybrid sources, reducing onboarding time by **50%** and ensuring repeatable, secure deployments.
- Built and optimized **data lakes (Amazon S3)** and **Redshift data warehouses**, enabling **40% faster querying** and supporting large-scale analytics for patient care metrics, utilization trends, and claims processing across business units.
- Enabled both **real-time and batch data processing** using **Apache Spark on EMR** and **Amazon Kinesis**, reducing ingestion latency by **30%** for critical data streams like lab results, vitals, and medication orders.
- Integrated **machine learning algorithms** and **Generative AI techniques** into healthcare data workflows by analyzing historical patient records to forecast readmissions and prioritize high-risk cases, achieving **92%+ predictive accuracy** and enhancing clinical decision-making across care teams.
- Developed **secure RESTful APIs** using **AWS API Gateway** and **Lambda** to expose curated datasets for internal analytics and ML workflows, supporting a **scalable, serverless data access layer** across data science teams.
- Built robust transformation and modeling pipelines using **Python**, **PySpark**, and **SQL**, ensuring **100% metric accuracy** and powering self-service BI for clinical, operational, and research stakeholders.
- Monitored and maintained pipeline health using **CloudWatch**, **Step Functions**, and **CloudTrail**, improving SLA adherence by **35%** while enforcing **enterprise-grade data governance (IAM, HIPAA)**; delivered **real-time Power BI dashboards** to operational leaders and execs.
- Implemented row-level security and **role-based access control** (RBAC) in Snowflake to ensure **HIPAA**-compliant data access across sensitive patient datasets.
- Developed reusable **DBT models** to standardize transformations and enable cross-functional analytics for clinical data across departments.
- Utilized **Azure Monitor** and **Terraform** to provision observability stacks for **Snowflake query** profiling and ETL performance dashboards.

ENVIRONMENT: AWS Glue, AWS Lambda, Amazon S3, Amazon Redshift, AWS EMR, Apache Spark, Apache Airflow, Amazon Kinesis, AWS CloudWatch, AWS Step Functions, Terraform, Python, PySpark, SQL, AWS API Gateway, IAM, AWS Access Analyzer, CloudTrail, Power BI

Ally Bank, Bangalore, India

Role: AWS Data Engineer **Sep 2021 - Jun 2023**

PROJECT: At Ally Bank, focused on building data-driven workflows to streamline auto loan pre-approvals, implementing targeted outreach, candidate filtering, and reminder automation to enhance approval rates and event engagement.

Responsibilities:

- Designed and deployed **scalable big data pipelines** using **Apache Spark, PySpark, AWS Glue, EMR, S3, Lambda**, and **SNS** to support Ally Bank's **Auto-loan Pre-Approval workflows**, reducing end-to-end data processing time by **30%**.
- Built **automated ETL workflows** using **AWS Glue Crawlers, Glue Studio**, and **Lambda**, enabling seamless ingestion from **S3, Snowflake, and transactional systems**, and reducing manual integration effort by **40%**.
- Optimized ETL transformations using **Glue DynamicFrames** and **Apache Spark**, improving processing efficiency by **20%**, while implementing **data enrichment** on dealership and customer event data for improved underwriting decisions.
- Administered and fine-tuned **SingleStore (MemSQL)** clusters using **Terraform, CloudFormation, and Ansible**, achieving **40% faster query execution** through optimized shard keys and indexing strategies.
- Integrated **Snowflake** as a centralized data warehouse, built **high-performance ELT pipelines**, and applied **clustering keys and virtual warehouse tuning**, reducing dashboard query response time by **40%** and improving reporting SLAs.
- Enforced **RBAC-based data governance** in Snowflake and AWS, ensuring **100% compliance** with internal banking policies and enabling secure cross-departmental data access via **secure data sharing**.

- Developed **Kafka-based data ingestion** in **Python**, supporting real-time streaming between microservices and enabling downstream analytics with near real-time customer behavior insights.
- Processed large-scale financial datasets in **Databricks** using **Python notebooks and Apache Spark**, enabling exploratory analysis and model training to support advanced customer segmentation and offer targeting.

ENVIRONMENT: Apache Spark, Python, SQL, AWS (EMR, S3, Glue, Lambda, SNS, Redshift), Snowflake, SingleStore (MemSQL), Apache Kafka, Databricks, Terraform, CloudFormation

SKF Group, Bangalore, India

Role: Data Engineer **Jun 2019 – Aug 2021**

PROJECT: At AB SKF, focused on modernizing manufacturing analytics by building secure, scalable pipelines on **GCP** to process equipment telemetry, inventory, and production data. Enabled real-time and batch data workflows to support predictive maintenance and operational reporting across global facilities.

Responsibilities:

- Migrated legacy data from **BDW Oracle** and **Teradata** into **Cloud Storage and BigQuery** using **Sqoop and Data Transfer Service**, while optimizing **PostgreSQL/SQL Server** structures for **30% faster queries**; also created interactive dashboards using **JavaScript, JSON, and AJAX** for operations and quality control teams.
- Built and maintained scalable **ETL pipelines** to process high-volume manufacturing data from supply chain and IoT systems using **Cloud Dataflow, Cloud Functions, Cloud Storage, and BigQuery**, improving throughput by **40%** and reducing manual touchpoints.
- Modelled production and inventory data using **Cloud SQL (PostgreSQL)** , and automated ingestion and transformation processes with **Python, Bash, and SQL**, cutting ETL cycle time by **60%**.
- Engineered **real-time data streaming pipelines** using **Pub/Sub**, **Apache Spark on Dataproc**, and **Cloud Composer**, enabling proactive monitoring of sensor telemetry and improving operational insights by **35%**.
- Implemented distributed data processing using **HDFS and YARN on GCP Dataproc**, scaling batch analytics for plant-wide performance metrics and energy efficiency models.
- Designed hybrid data integration pipelines using **Apache NiFi**, **Talend**, and **Cloud Data Fusion** to consolidate data from **ERP and inventory systems** into **GCP data lakes** on **Cloud Storage** and **BigQuery**.
- Enabled low-latency data streaming and operational dashboards using **Kafka (on Confluent Cloud)**, **Zookeeper**, **Apache Flume**, and **BigQuery with BI Engine**, reducing alert latency by **45%** for factory floor events.
- Developed **predictive ML models** using **TensorFlow, Scikit-learn**, and **Vertex AI** to forecast machinery failure and optimize production schedules, achieving **25% reduction** in downtime and better inventory utilization.

ENVIRONMENT: Cloud Data Fusion, Apache NiFi, Talend, Google BigQuery, Cloud Storage, Apache Spark, Dataproc, Cloud Composer, Pub/Sub, Kafka, Snowflake, TensorFlow, Vertex AI, Scikit-learn, Sqoop, Oracle, Teradata, PostgreSQL, SQL Server, Python, Bash, Airflow, Stackdriver, IAM, RBAC

TECHNICAL SKILLS

Programming Languages	Python (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn), Java, C#, C++, R, SQL, Hive, Spark, PowerShell
Cloud Platforms	AWS, Azure, GCP
Data Visualization Tools	Tableau, Power BI, Looker, QlikView, D3.js, R(ggplot2), Python ((Matplotlib, Seaborn)
Hadoop Ecosystem	Apache Hadoop, Spark, Snowflake, MapReduce, HiveQL, HDFS, Sqoop, PigLatin, Apache Kafka, Flink, Storm, HBase
Databases / Data Sources	PostgreSQL, Microsoft SQL Server, MongoDB, MySQL, IBM DB2, Oracle, PL/SQL, DynamoDB, HBase, Amazon Redshift, Teradata, RDBMS, Snowflake, Cassandra
Version Control & CI/CD	Git, GitHub, Bitbucket, Jenkins, Maven
Containerization & Orchestration	Docker, Kubernetes, Apache Mesos, DBT, Apache Airflow
Data Modeling & Design	Star Schema, Snowflake Schema, Dimensional Modeling, Data Vault, OLAP, OLTP
Data Warehousing	Amazon Redshift, Google BigQuery, Snowflake, Azure Synapse Analytics, Teradata, Vertica

Operating Systems & Methodologies	Agile, Scrum, Kanban, Waterfall, UNIX Shell Scripting (via PuTTYclient), Linux, Windows, MacOS
Operating Systems & Development Tools	Linux, MacOS, Windows, Unix, IntelliJ IDEA, Eclipse, Visual Studio Code
ETL Tools	Apache Nifi, Apache Airflow, Informatica, SSIS

EDUCATION

Master’s in business Analytics, University of Cincinnati, Ohio, USA Dec 2024